

OMNIVERSAL ARCHITECT LABS

OPEN WINDOW DOCTRINE · CONDITIONAL FAILURE AT THE ROOT LEVEL · PRESENTATION-STATE INTEGRITY

TEMPORAL ALIASING INCORPORATION FRAMEWORK

Independent Development Pass — Claude

A research account, incorporation map, dual-domain simulation-lab action plan, funding pathway, and critical-path analysis for reconciler mismatch across mind, device, and nested system.

Working draft for Architect review. Not canon. Simulation-only safety posture. Benign engineered proxies only.

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Document Purpose, In One Sentence

This pass gives temporal aliasing a research-grade, source-checked account; places it inside the locked Open Window v0.3 architecture and the Conditional Failure at the Root Level (CFTRL) / Presentation-State Integrity Framework (PSIF) bodies of work; and specifies a dual-domain (simulation-plus-real-world) lab program, funding pathway, and critical-path list — while flagging, rather than smoothing over, the two places the underlying science is contested and the two places this pass disagrees with the prior draft.

1. Mission and Purpose

1.1 Mission

The mission is to take a lived observation — that a brain, and every device it is coupled to, each reconstruct a usable “now” from incomplete, differently clocked input — and turn it into a falsifiable, buildable engineering object, without inflating every mismatch into either “everything is an illusion” or “something got in.” Temporal aliasing is the name for one specific, well-understood class of that reconstruction error: when a system samples a process more slowly than twice the highest frequency present in it, the high-frequency structure folds into a lower-frequency impostor. This is not a metaphor borrowed for flavor — it is a proven theorem (Nyquist–Shannon) with a public demonstration everyone already recognizes: the wagon-wheel effect.

The mission is explicitly not to prove that every timing mismatch is deception, coordination, or “infection.” It is to measure how often a presented timeline diverges from an evidenced state-transition ledger, under what sampling and synchronization conditions that divergence becomes systematic and reproducible, and whether the system that produced it can name and correct the divergence or can only keep serving the aliased version after the gap is shown to it.

1.2 Purpose of this document

- Give the Architect a research-grade, cite-checked account of temporal aliasing and the neighboring family of temporal phenomena — accurate enough that terms don't get flattened into each other later.
- State exactly how temporal aliasing sits inside CFTRL / the Department of Endings, the Open Window Doctrine (CETE, optional airlock, declared proxy classes), and PSIF (presentation vs. state, the evidence ladder, the failure modes).
- Specify a lab program — simulation and real-world — that a small team could actually stand up, staged by gate rather than by calendar.
- Provide a funding pathway, a critical-implementation-items list, an alternative analysis, and an explicit questions/comments/concerns section.
- Say plainly what works, what does not, and what is still unaddressed — including where this pass disagrees with the existing v0.1-GROK draft on the same topic.
- Keep ONESTAMP authority with the Architect. Nothing here is a finding about any person, organization, or incident.

1.3 Why this is not optional, stated plainly

Every device in a person's life — phone, watch, television, laptop, car dashboard — is a clocked sampler. The nervous system using those devices is also a clocked sampler, with its own effective bandwidth, its own latency, and its own willingness to invent smooth continuity where the data is ambiguous. What a person calls “what actually happened” is the reconciled output of that whole stack, not a readout of a single ledger. Treating the reconciled output as if it were the ledger is the exact error PSIF calls presentation capture. Treating every mismatch as proof of hidden motive is the opposite error, attribution inflation. Temporal aliasing is one of the cheapest, most reproducible ways to manufacture both errors at once, because it produces a confident, smooth, wrong answer rather than an obviously broken one.

2. Executive Summary

Temporal aliasing is the reconstruction of a false lower-frequency (and sometimes reversed-direction) identity from a signal sampled too slowly. This is settled signal-processing theory, not speculation: the Nyquist–Shannon sampling theorem states that a signal with no energy at or above B Hz can be exactly reconstructed from uniform samples only if

the sampling rate exceeds 2B; below that rate, information loss is not partial, it is a categorical substitution of one frequency identity for another, and it cannot be recovered after the fact by looking harder at the same samples.

The filmed wagon-wheel effect — a spoked wheel that appears to slow, stop, or spin backward on camera — is the textbook public demonstration, documented since Purves, Paydarfar & Andrews (1996) and used in vision-science and flight-simulator display literature ever since. A second, more contested phenomenon, the continuous wagon-wheel illusion (observers reporting reversal under ordinary continuous light, no camera involved), has at least two competing explanations in the literature — a discrete-attentional-sampling account and a motion-detector-rivalry account — and this pass treats that dispute as a live scientific question, not a settled fact to build doctrine on.

The Architect's observation generalizes the aliasing theorem across three stacked reconcilers that do not share a clock: (1) the nervous system, which binds, delays, and — per the postdiction literature — sometimes revises what is attributed to a moment up to roughly 80 milliseconds after it, though a competing differential-latency account explains some of the same illusions without invoking retroactive rewriting; (2) each device, independently sampling, buffering, and time-stamping; and (3) any organizational or nested-simulation process that must produce one coherent timeline from those streams. None of the three is the root clock. When the root — a maintenance process, a reference time source, an original event — becomes unreachable, drifted, or abandoned (Conditional Failure at the Root Level), the stack does not stop producing a picture. It aliases, smoothly and without complaint.

This pass frames temporal aliasing as, simultaneously: a Cross-Environment Transfer Event (CETE) subtype under Open Window v0.3 when a timing, order, or duration effect crosses a declared interface; a presentation-state divergence under PSIF when the displayed “now,” order, or duration disagrees with an evidenced ledger; a CFTRL failure mode when a lower layer cannot inspect or correct the sampling policy of the layer above it; and a Department of Endings work item — every temporal failure mode (fold, freeze, reversal, stitch, drop, postdictive rewrite) is run, logged, and mined for a lesson rather than treated as noise.

The recommended first executable program is a four-card MVP using only declared interfaces and benign engineered proxies (injected timestamp offsets, declared sample-rate deficits, controlled clock skew, and scripted content-timing offsets), plus one dual-channel real-world observational protocol that never claims a universal ground-truth clock outside the simulator. Funding is staged by capability, not by date: a lean desktop-and-Python prototype is achievable in roughly the \$10,000–\$30,000 / 80–150 person-hour range; a mid-tier instrumented lab with independently sourced clocks sits around \$35,000–\$85,000; a high-fidelity multi-device chamber with hardware timestamping runs \$150,000 or more and is explicitly not required to clear the first four gates.

The single largest implementation risk is category collapse — calling every timing oddity “aliasing,” every aliasing event a “transfer,” and every transfer “infection.” This pass follows PSIF's own discipline against that collapse: occurrence, attribution, causation, and intent are graded and reported separately, and a null result (no material divergence found) is written up as a real result, not a failed one.

3. Overview and Detailed Research Account

3.1 Formal definition and why it is not a metaphor

In signal processing, aliasing is what happens when a continuous signal is sampled below its own information rate: components at or above the Nyquist frequency (half the sampling rate, $f_s/2$) become indistinguishable, after sampling, from lower-frequency components. A signal at frequency f sampled at rate f_s produces the same sample sequence as a signal at $|f - k \cdot f_s|$ for whatever integer k brings the result into the representable range. Below the Nyquist rate the loss is not graceful degradation; the original identity is not recoverable from the samples alone, which is exactly why a pre-sampling anti-alias filter (removing energy above $f_s/2$ before sampling) is the only real defense — “look more closely at the log” cannot undo it after the fact.

Grounding: Nyquist–Shannon sampling theorem as treated in standard DSP references (e.g., DSPRelated Foundations series; Fiveable EE course notes); U.S. Patent literature on temporal aliasing in motion-picture capture describes the identical fold mathematically for a sinusoidally driven light source sampled by a 24 fps camera.

3.2 The wagon-wheel family

The classic, camera-shutter wagon-wheel effect is temporal aliasing applied to a periodic spatial pattern: if a wheel advances close to, or more than, half a spoke-interval between frames, the nearest-neighbor match the visual system (or a video codec) settles on can be the wrong direction. Purves, Paydarfar & Andrews (PNAS, 1996) formalized this as the

public-facing demonstration of spatio-temporal aliasing in sampled imagery, a treatment later extended in flight-simulator and display-engineering literature and in follow-up psychophysics work (Finlay & Dodwell 1987; Martineau, Aguilar & Glass 2009 on predicting perceived direction).

The continuous wagon-wheel illusion (c-WWI) is a different, contested claim: some observers report apparent reversal under steady, non-strobed illumination. One line of work treats this as evidence that visual motion perception is itself discretely sampled, in the neighborhood of 10–15 Hz; a competing line explains the same reports as rivalry between opposed-direction motion detectors, with no global sampling clock required, and delay-coupled neural-field models have since reproduced quantized apparent speeds and reversals from axonal delay alone. This pass treats the dispute as useful rather than inconvenient: a lab program built on this material must not assume a single brain frame-rate, and should treat “does this observer or this device chain alias under continuous drive?” as an empirical, individual- and condition-dependent variable rather than a constant to plug into a formula.

3.3 Neighboring temporal phenomena — kept deliberately distinct

Aliasing is one member of a family of timing effects that are easy to blur together in casual description but that have different mechanisms, different evidence requirements, and different lab relevance. Collapsing them into one word (“glitch,” “lag,” “aliasing”) is the single fastest way to wreck a metric dictionary and make every subsequent result unfalsifiable.

Phenomenon	What it is	What it is not	Lab relevance
Temporal aliasing	High-frequency structure reconstructed as a lower-frequency (possibly reversed) impostor because $f_s \leq 2B$	Any delay; any illusion in general; a deliberate lie	Primary independent variable: declared sample rate vs. declared event rate
Clock skew / drift	Two clocks disagree on the value, and possibly the rate, of the same instant	Aliasing; a merely slow clock is not a folded frequency	Primary real-world confounder; requires NTP/PTP-class discipline to bound
Causality / order violation	Observed order disagrees with a happens-before relation, even with reasonable clocks	Proof of intent or coordination	Order-inversion CETE subtype; compare against Lamport/vector/hybrid logical clocks
Jitter / delay variance	Sample intervals or latencies are uneven	A change in mean rate	Breaks the uniform-sampling assumption the Nyquist theorem relies on
Buffering / batching	Events are held and released as a clump	Sampling below Nyquist	Produces visible “catch-up” stitches, not folds
Drop / freeze	A sample is omitted, or a frame is held static	Direction reversal specifically	Distinguish “nothing happened” (M0) from “a frame is being reused as if it were new data”
Postdiction (contested mechanism)	Perception attributed to time t_0 can be revised by input arriving afterward, in paradigms such as flash-lag	Forward prediction; a settled, uncontested account	A competing differential-latency account (Patel, Ögmen, Bedell & Sampath 2000) explains much of the same flash-lag data without retroactive rewriting; the lab must not assert postdiction as fact, only as one live hypothesis
Temporal binding window	Asynchronous multimodal events are perceived as one event if they fall inside a window on the order of tens to a few hundred ms	Proof the events actually co-occurred in a ledger	Device-plus-body stacks routinely bind events a ledger would keep separate

Phenomenon	What it is	What it is not	Lab relevance
Chronostasis (“stopped clock” effect)	The first percept after a saccade or attention shift is reported as lasting longer than it did	A form of aliasing	Occupant duration reports are systematically biased at attention shifts — a confound for any human-report metric
Critical flicker fusion (CFF)	Flicker above a threshold is perceived as steady light; commonly cited in the tens-of-Hz range but strongly condition-dependent	A fixed, universal “brain frame-rate”	Any claim that treats 60 Hz (or any single number) as the brain's refresh rate is a category error to guard against explicitly

3.4 The brain as an imperfect reconciler — stated without romance or contempt

The empirical record supports the Architect's framing closely, and it should be stated in plain systems language rather than mystified: the nervous system does the best it can with incomplete, asynchronous, differently latent input streams, and what reaches awareness is a bound, delayed, prior-shaped composite — not a transcript. Several documented limits matter directly to a lab program built on this idea:

- Flicker fusion is not one number. Reported thresholds move with luminance, stimulus size, retinal location, fatigue, age, and substances; “the brain runs at 60 Hz” is a false and unhelpfully precise claim that should be excluded from any lab document as doctrine.
- Attentional-sampling accounts of c-WWI cluster near 10–15 Hz, but a rivalry/delay-based account denies a global shutter exists at all; expect real inter-observer variance rather than a shared constant.
- Postdiction (Eagleman & Sejnowski, 2000) proposes that the percept assigned to a moment can depend on roughly 80 ms of subsequent input, based on the flash-lag effect; a published, peer-reviewed rebuttal (Patel, Ögmen, Bedell & Sampath, Science, 2000) argues the same data is fully explained by differential neural latency between a moving and a flashed stimulus, with no retroactive rewriting required. Later work (e.g., generalized and cross-modal flash-lag studies) finds evidence for both latency-difference and integration-window mechanisms depending on the paradigm. This pass therefore reports postdiction as a live, contested hypothesis with real empirical support, not as an established mechanism, and it recommends the lab program stay agnostic about why an occupant's timing report diverges from the ledger and simply measure that it diverges, and by how much.
- Temporal binding windows vary by modality pair, stimulus intensity, age, and task; events a ledger separates by 50–200 ms are routinely experienced as a single event.
- Optical and temporal illusions exist because perceptual reconstruction is a prior-driven inference process, not a passive recording. Illusion demonstrations are useful for building shared intuition; they are not, by themselves, evidence that any particular device, aperture, or interface in someone's home is compromised, and a lab document should never cite a YouTube illusion demonstration as validating evidence for a specific real-world claim.

The compact systems sentence for this section: the brain is a lossy, delayed, prior-using reconciler whose single output is treated by its owner as root-level truth, precisely because the owner has no layer above their own awareness from which to audit it. That is Conditional Failure at the Root Level, restated in the first person, and it is the cleanest bridge between the neuroscience and the existing OAL framework.

3.5 The device stack as a second reconciler

Every device a person carries or is surrounded by runs its own independent sampling-and-presentation policy, and the failure modes are well documented in systems and distributed-computing practice rather than novel to this framework.

Layer	Typical timing object	Common failure / alias mode
Sensor / camera / touch	Frame time; integration window	Motion blur; rolling-shutter skew; undersampled gesture
OS / app clock	CLOCK_REALTIME vs.	NTP step; backward time jump; wall-vs-

Layer	Typical timing object	Common failure / alias mode
	CLOCK_MONOTONIC	monotonic disagreement
Display	Refresh rate (60/90/120 Hz); VSYNC; frame queue	Tearing; held frames; wagon-wheel artifacts on UI motion
Media pipeline	Encoded GOP structure; jitter buffer	Audio/video desync; sudden “catch-up”; dropped group-of-pictures
Network	Round-trip time; bufferbloat; NTP/PTP path	Reordering; burst delay; reference-stratum failure
Cross-device / cloud	Server timestamp; last-write-wins merge	A stale write silently wins; a reply appears to precede its question

Physical-clock disagreement is a routine enough hazard in distributed systems that the field simply refuses to treat wall-clock order as causal order — Lamport timestamps, vector clocks, and hybrid logical clocks all exist specifically because physical clocks disagree just enough to invert happens-before relationships. An occupant who treats “what my phone and my television agree happened” as an authoritative ledger is making, informally and in a living room, the same last-write-wins mistake distributed-systems engineers spent decades learning to design around.

3.6 Why reliance on technology enlarges the gap, not just the convenience

Device dependence does not add a single new reconciler on top of the brain; it multiplies reconcilers, and each additional device is itself trying to reconcile against a network and against other devices. The composed presentation a person trusts can be locally self-consistent and still be globally wrong about order, duration, or cause. That is the concrete, non-speculative reason this belongs in a systems-integrity and security-posture lab rather than only in a philosophical discussion: it is directly relevant to incident review, safety investigation, multi-device UX design, simulation fidelity, and any protocol that treats a timestamp as if it were a causal arrow.

4. Incorporation Map Across Existing OAL Bodies of Work

4.1 Conditional Failure at the Root Level (CFTRL)

CFTRL's central question — what happens when a stacked system is scaled and the root fails, and lower tiers cannot climb up to repair it — has a direct, literal answer in the timing domain. If a root clock, root sampling policy, or root maintenance process becomes unreachable, drifted, or abandoned, every layer beneath it keeps producing a coherent-looking picture; that picture is an alias of a process the layer can no longer inspect. The wishbone brittleness image from OAL-CFTRL-NAIL-001 has a direct timing analogue: two clocks corrected in opposing directions, or a sampling grid too coarse for the true event rate, both “work” at small scale and fold catastrophically past a threshold rather than degrading gracefully.

4.2 The Department of Endings

As CFTRL's operational arm, the Department of Endings gains a natural Temporal Endgame function in this incorporation: run every named temporal failure mode (fold, freeze, reversal, stitch, drop, postdictive rewrite) through the stack under controlled conditions, log the outcome, extract the lesson, and feed it back — treating no failed run as wasted. Staffing, character, and institutional-form questions from OAL-CFTRL-NAIL-001 remain open author-input items and are not resolved by this pass.

4.3 Open Window Doctrine (v0.3 governing baseline)

Under the locked v0.3 decisions, “window,” “aperture,” and “interface” are provisional functional terms for any vector across which observation, state, timing, or influence may cross — not limited to screens — and the Interface Airlock is an optional control, not an assumed property. This incorporation adds one clean requirement: an aperture's declared state should include a timing profile (sample rate, buffer depth, clock source, last synchronization quality), because a powered-but-dark display, a paused media player, and a notification radio that still wakes a processor are three different timing states even though v0.3 already correctly refuses to treat them as one binary “on/off.”

Cross-Environment Transfer Event (CETE) remains the parent term for any verified crossing; temporal aliasing is a labeled subtype (a “fold” — high-rate structure reconstructed as low-rate or reversed) alongside neighboring, distinct subtypes such as order inversion, duration distortion, clock/stratum skew faults, and buffered stitch-release. “Infection” stays available only for an unwanted, propagative subtype and is never the default label for an ordinary timing fold.

4.4 Presentation-State Integrity Framework (PSIF)

PSIF's core question — do an environment's claims correspond to its evidenced operational state and causal structure — applies to time exactly as it applies to any other claim. “This happened now,” “this happened first,” and “this lasted three seconds” are presentations, not automatically state, and the correct audit method is provenance-based reconstruction against an independent ledger with occurrence, attribution, causation, and intent graded and reported separately (PSIF's evidence ladder, Section 5.1 of the PSIF Independent Development document). The PSIF distinction between event fidelity and causal fidelity is exactly the right tool here: a system can display every individual timestamp accurately and still misrepresent the order, grouping, or duration relationships among them — which is causal masking performed by a clock stack rather than by a narrator, and it is worth stating that this can happen with zero intent behind it at all.

The Copilot Independent Pass on PSIF adds concepts this lab program needs specifically after a divergence is detected: Correctability (can the operator actually fix the sampling or clock policy?), Presentational Inertia (does the environment keep serving the aliased timeline even after the gap is demonstrated?), and the Assertion Boundary (do not state intent at a strength the evidence supports). A temporal-aliasing finding that a system “acknowledges a glitch” while continuing to serve the reconciled, aliased story to the user is, in PSIF's own vocabulary, a Level-1 inertia case — worth naming precisely rather than folding into a generic complaint.

5. Simulation-Lab and Real-World Action Plan

5.1 Design principles

- Flexible, exit-criteria-driven timing (per the locked v0.3 Decision 1): no scenario is treated as a success because a calendar window closed. Advancement is gated (see 5.4).
- Declared interfaces and benign engineered proxies only for the MVP. No undeclared-vector hunting; a surprising result is logged as a Candidate Interface Event and held, not promoted (per v0.3 Decision 2 and 3).
- Every MVP card exists in both a simulation form (with a canonical engine ledger) and a real-world analogue (with at least two independent timing channels and no claim of a universal ground-truth clock). A single-channel real-world run is instrumentation-limited and cannot support a temporal-aliasing claim, full stop.
- The simulator's audit clock and its device-layer clock must not share a crystal, NTP path, or random seed. Shared timing infrastructure between the “ledger” and the “device under test” produces a sensor-agreement illusion — an artifact that looks like confirmation and is actually just one clock agreeing with itself.

5.2 A minimal, working layer model for timing work

Layer	Name	Timing object
L0	Audit / reference plane	Independent reference clock; append-only, hash-chained event ledger
L1	Canonical engine state	Simulation tick; authoritative event and dependency IDs
L2	Environment process	Physical or simulated dynamics at their native rate
L3	Device / aperture object	Per-device sample rate, buffer depth, clock source
L4	Occupant reconciler	Binding window, postdiction-or-latency window, flicker-fusion threshold, attention state
L5	Narrative / HUD claim	The presented order, duration, and “now” actually shown to the occupant

5.3 MVP scenario cards (four, plus one real-world protocol)

These cards are compatible with, and deliberately do not duplicate, the four TA cards already drafted in OAL-TA-001 v0.1-GROK (undersampled motion, multi-device “now,” black-screen residual clock, and benign content-timing bleed). This pass recommends running that existing card set as written, with one addition and one modification flagged below.

Addition — TA-05: Postdiction-vs-Latency Discriminator

Purpose: because Section 3.4 identifies postdiction as a contested mechanism rather than settled fact, this card exists specifically to keep the lab from silently adopting one side of an open scientific debate. Design a flash-lag-style probe with two conditions that the postdiction and differential-latency accounts predict differently (following the logic in Patel, Ögmen, Bedell & Sampath 2000), run it against the ledger, and report which account the collected data favors — or report that the data cannot distinguish them, which is itself a valid and useful result. Metrics: perceived-vs-ledger offset by condition; a written statement of which theory the result supports, if either.

Modification — TA-02 (Multi-Device Now): add an explicit human-report abstention option

The existing card asks an occupant or operator to flag simultaneity of two signals. Because chronostasis and attention shifts are known to distort human duration reports independent of any device-timing effect (Section 3.4), the card should let a reporter register “uncertain” rather than forcing a binary simultaneous/not-simultaneous call, and should log attention-shift events (saccades, alerts) as a covariate rather than discarding them.

Real-world observational protocol (RW-01): Household Multi-Aperture Timing Audit

A fully non-invasive, device-only protocol that never touches occupant awareness claims. Instrument a small set of consumer devices already in ordinary use (phone, TV, smart-speaker, router) with a passive, isolated-network time-logging tap. Record each device's self-reported clock, its NTP/PTP stratum where exposed, and timestamps on a shared benign event (e.g., a scripted on-screen countdown). Compare cross-device skew and drift over a multi-day observation window against a reference time service. This protocol produces exactly the kind of evidence Section 3.6 calls for — a documented, non-metaphysical account of how much two “agreeing” devices in one room can actually disagree — without any claim about occupant perception, consent-sensitive content capture, or anything resembling surveillance of a person rather than of declared, lab-owned hardware.

5.4 Gate structure (advancement, not calendar, decides)

Gate	Question	If it fails
G0	Interfaces, proxy class, and airlock condition declared in advance; null hypothesis written?	Block the run; fix the card
G1	Independent timing channel present and hashed/tamper-evident?	Block; fix instrumentation before proceeding
G2	Positive and negative controls behave as specified?	Revise the card
G3	Primary metric moved in the predicted direction under the declared manipulation?	Report inconclusive or falsified — do not relabel
G4	Replay on stored raw traces reproduces the effect?	Candidate finding only; not doctrine
G5	Independent recurrence on a fresh run?	Hold any promotion to canon
G6	After-action review and exception log complete; review-group recommendation written?	No ONESTAMP eligibility

5.5 Safety and ethics constraints

- Simulation-only for any language about occupancy, awareness, or “infection.” The real-world protocol is observational and device-timing only, never a claim about a person's mind.

- No biological-harm framing anywhere in the timing work, and no undeclared stimulation of any human participant.
- No “partially unaware operator” condition in the MVP under any framing; this remains an unresolved ethics item, out of scope until a dedicated protocol exists.
- If any human timing judgment is collected at all, it requires informed consent, the right to withdraw, and no deception beyond the specific, pre-registered illusion stimulus being tested. Device telemetry is strongly preferred over occupant interrogation for the MVP.
- A pre-registered stop condition (maximum tolerated clock skew for the platform) halts any run automatically; an undeclared/unexpected vector is logged as a Candidate Interface Event and never promoted without independent recurrence.

6. Funding Pathway

Funding should track capability gates, not a fixed calendar, consistent with the locked v0.3 timing decision. Figures below are planning estimates for a small, largely internal team using consumer and light lab-grade hardware; they are not vendor quotes.

Tier	Scope	Rough cost band	Effort	Gates it can clear
Tier 0 — Desktop / Python prototype	T0 timing-stack simulator (Section 5.2 layer model) with injected fold, skew, and root-maintenance-kill conditions; no physical hardware beyond existing computers	\$10,000–\$30,000 equivalent	~80–150 hours	G0–G4 achievable entirely in software
Tier 1 — Instrumented device lab	Independent reference clock, isolated test SSID/network, 3–5 consumer devices, packet-capture and logging pipeline, basic hardware timestamping	\$35,000–\$85,000	~200–400 hours	G1–G5 with real hardware; supports RW-01
Tier 2 — High-fidelity multi-device chamber	Hardware-timestamped multi-device rig (PTP-class), controlled lighting/display array for wagon-wheel and c-WWI style visual work, dedicated review-group cadence	\$150,000–\$280,000+	Ongoing program	Full G0–G6 with independent recurrence; not required before Tier 0/1 results exist

Recommended sequencing: fund Tier 0 first and treat it as a gating deliverable in itself. A Tier 0 result that reproduces the wagon-wheel fold and a documented cross-device skew distribution, replayed and independently recurring, is a legitimate, presentable proof of concept that can justify Tier 1 funding without requiring any physical lab spend up front. Tier 2 should not be funded speculatively; it exists to answer questions Tier 1 raises, not to replace Tier 1.

7. Critical Items That Need to Be Addressed Before Implementation

- ONESTAMP disposition of this pass and of v0.1-GROK: are both retained as parallel independent passes feeding a later synthesis, or does the Architect want one merged v0.2 immediately? (Recommendation: keep them parallel until Tier 0 data exists — merging two draft frameworks before any data exists risks locking untested assumptions.)
- Review-group membership and ONESTAMP trigger language, referenced in the Open Window Implementation Memo, is still an open governance item and blocks any run from being promotable even if it succeeds technically.
- A single, one-page metric dictionary must be circulated to every AI collaborator before any card is run, so “skew,” “alias,” “fold,” and “divergence” mean the same thing across ChatGPT, Grok, Gemini, Copilot, Perplexity, and this pass. Right now the definitions are correct but scattered across four-plus documents.

- A decision on whether PSIF's Systemic Presentation-Layer Deception (SPLD) umbrella covers temporal cases by default, or whether “temporal presentation–state divergence” is carved out as a named, non-deceptive default category. This pass's recommendation (Section 9) is to carve it out explicitly, because aliasing can be entirely innocent and still operationally significant, and folding it under a deception-flavored umbrella term invites exactly the attribution-inflation error PSIF exists to prevent.
- A seeded, version-controlled repository for the T0 Python model with a running defect log, so results are reproducible by any collaborator rather than living only in chat history.
- Explicit retirement of superseded file names/versions across the Open Window family so a future AI pass does not accidentally synthesize a “ghost” draft from stale filenames.
- A short, plain-language consent sheet, ready in advance, for the one case (TA-05 / postdiction discriminator) where any human timing report is collected at all.

8. Alternative Analysis

8.1 Alternative framing: skip the neuroscience, keep only the systems claim

A leaner version of this whole framework is possible: drop every claim about brains (postdiction, chronostasis, flicker fusion) and keep only the device-to-device systems claim — clocks disagree, sampling below Nyquist folds, distributed systems already have a 40-year literature on this. This alternative is faster to fund and immune to the open neuroscience disputes in Section 3.4. Trade-off: it loses the part of the Architect's original observation that motivated the whole document — the parallel between a brain reconciling sensory streams and a device reconciling data streams — and it would need to be reintroduced later anyway once the lab wants to test occupant-facing effects (TA-02, TA-04, TA-05).

8.2 Alternative framing: treat this as pure PSIF, no new CETE subtype

Instead of adding CETE-TA as a new subtype, timing divergence could simply be logged as an ordinary PSIF state-claim divergence with a “time” tag, with no Open-Window-specific vocabulary at all. This is simpler and avoids adding taxonomy. Trade-off: it loses the Open Window–specific machinery (declared apertures, airlocks, aperture timing profiles) that is genuinely useful for the multi-device case, and it would make cross-referencing this work against the rest of the Open Window canon harder rather than easier.

8.3 Alternative framing: skip simulation, go straight to the real-world protocol

Because RW-01 is non-invasive and can run on existing consumer hardware, one could skip Tier 0 entirely and start with real-world device auditing. Trade-off: without a canonical-ledger simulation first, there is no independently verified ground truth to calibrate the real-world instrumentation against, and any real-world “finding” would be a single-channel, unfalsifiable claim exactly of the kind Section 5.1 rules out. This pass recommends against this ordering.

8.4 Where this pass disagrees with v0.1-GROK

- v0.1-GROK presents postdiction largely as an accepted mechanism with a caveat about c-WWI attentional sampling being contested. This pass treats postdiction itself as contested (a real, published differential-latency alternative exists for the same flash-lag data) and adds TA-05 specifically to keep the lab agnostic rather than adopting one side.
- v0.1-GROK's funding bands (\$12,000–\$35,000 / \$40,000–\$90,000 / \$150,000–\$280,000) and this pass's bands are close but not identical; this pass's Tier 0 floor is slightly lower because a Python-only reproduction of the fold and skew phenomena genuinely requires no lab hardware at all, only compute time already available.

9. Questions, Comments, and Concerns

9.1 Questions for the Architect

- Should “temporal presentation–state divergence” be carved out of SPLD as a named non-deceptive default (this pass's recommendation), or should it stay folded under the existing SPLD umbrella for simplicity?
- Is a small, real, consent-sensitive human-timing-report component (TA-05) something the Architect wants funded now, or should the MVP stay device-telemetry-only until a dedicated human-subjects protocol exists?

- Does the Architect want this pass and v0.1-GROK kept as two parallel independent drafts pending Tier 0 data (this pass's recommendation), or merged into a single v0.2 before any run happens?

9.2 Comments

The underlying physics and signal-processing claims in both this pass and v0.1-GROK are solid and well-sourced; the risk in this project is not the science, it is documentation sprawl. Four to six substantial documents (Open Window v0.1–v3.0, the Decision Record, the Implementation Memo, PSIF's two independent passes, OAL-CFTRL-NAIL-001, and now two TA passes) already exist before a single scenario card has actually been run. The most valuable near-term action is probably not another framework document — it is standing up Tier 0 and producing one real, replayed, independently-recurring result.

9.3 Concerns

- Confirmation bias: the framing already “expects” devices and reconcilers to fold in interesting ways. The null-hypothesis and gate structure in Section 5.4 exist specifically to let a boring, no-effect result win, and that structure needs to actually be honored when a card produces a null.
- Sensor-agreement illusion: if the audit ledger and the device under test ever share a clock source, a “confirmed” result is an artifact, not a finding. This is the single easiest mistake to make by accident when building the T0 prototype quickly.
- Category collapse under narrative pressure: once a real fold is demonstrated once, there will be a natural pull toward calling every subsequent odd timing event “aliasing” or “CETE-TA” without re-running the gate structure. The metric dictionary (Section 7) is the main defense.
- Scope creep the moment a card “looks cool”: TA-01 (undersampled motion) is the most visually compelling card and the most likely to attract premature expansion before G1–G4 are actually satisfied.

10. What Works, What Does Not, and What Else Must Be Addressed

10.1 What works

- The locked Open Window v0.3 decisions are a sound baseline for this incorporation; nothing here requires revisiting them.
- CETE as the parent transfer term lets timing effects be described without borrowing biological or metaphysical language by default.
- PSIF's evidence ladder and assertion boundary are the correct tools for keeping occurrence, attribution, causation, and intent separate — this is the single most important piece of existing machinery for this whole project.
- CFTRL plus the Department of Endings gives temporal failure modes a natural home and a reason to log rather than discard null and negative results.
- A gate-driven, not calendar-driven, advancement rule protects the project from declaring victory on a fixed schedule.

10.2 What does not work

- Treating “the brain runs at X Hz” as a settled number for any X — the literature does not support a single constant, and using one would undermine the credibility of every downstream claim.
- Adopting postdiction as an established mechanism rather than a contested hypothesis with a real published alternative.
- Using “infection” as a default label for ordinary timing bleed; it should remain reserved for a genuinely unwanted, propagative subtype.
- Running any card without an independent, non-shared timing channel — this produces confirmation, not evidence.

- Building a fourth or fifth “final, unified” Open Window merge document before any lab data exists; this document is deliberately not attempting to be that.

10.3 What else must be addressed

- The still-open Open Window governance questions (review-group membership, pause authority, stop thresholds, artifact home for new documents) block promotion of any result regardless of how clean the data is.
- A circulated, single-page metric dictionary, not one buried inside a long draft, is needed before multiple AI collaborators start producing inconsistent labels for the same measured quantity.
- A decision on the SPLD-vs-carve-out question (Section 7) should happen before, not after, the first real divergence is found, so the finding is not shoehorned into whichever category is momentarily convenient.

11. Immediate Next-Step Action Plan

- Architect review and mark-up of this pass alongside v0.1-GROK: accept, revise, or reject the CETE-TA vocabulary, the layer model, and the TA-05 addition.
- Decide the SPLD carve-out question (Section 7) and record the decision as a short addendum, not a new unified doctrine rewrite.
- Stand up the Tier 0 Python timing-stack prototype with injected fold, skew, and root-maintenance-kill conditions; this can begin immediately and requires no funding decision.
- Lock review-group and ONESTAMP trigger language from the Open Window Implementation Memo before any card is treated as a live run rather than a rehearsal.
- Run the existing TA-01 through TA-04 cards (v0.1-GROK) plus TA-05 (this pass) under the Section 5.4 gate structure; preserve raw traces; replay before any claim is written up.
- After-action classification of each card as supported, inconclusive, falsified, blocked, or deferred — including publishing null results with the same seriousness as positive ones.
- Only after a Tier 0 result clears G4–G5 should Tier 1 funding or RW-01 field deployment be considered.

12. Working Vocabulary for This Pass

Term	Meaning in this document
Temporal aliasing	Reconstruction of a false lower-frequency (possibly reversed) identity from a signal sampled below twice its highest present frequency
Nyquist frequency	Half the sampling rate ($f_s/2$); the folding boundary above which aliasing occurs
Reconciler	Any process — nervous system, device, or organization — that emits one timeline from multiple, differently clocked input streams
CETE-TA	Cross-Environment Transfer Event whose observable effect is a timing fold
Canonical ledger	An append-only, hash-chained state-transition record used as simulation ground truth; not assumed to exist in real-world settings
Postdiction (contested)	Hypothesis that a percept attributed to a moment can be revised by input arriving afterward; treated here as one live hypothesis among competing accounts, not settled fact
Presentational inertia (timing)	An environment continuing to serve an aliased timeline after the divergence has been demonstrated to it

Authentication Block

Document ID	OAL-TA-001-CLAUDE
Version	1.0-CLAUDE
Date compiled	7 September 2026
Compiled by	Claude (research / synthesis / independent audit layer)
Architect of Record	Nathan Lewis II
ONESTAMP	PENDING — not applied
Relationship to prior TA draft	Parallel independent pass to OAL-TA-001 v0.1-GROK; not a supersession
Canon status	DRAFT — UNDER REVIEW
Safety posture	Simulation-only for occupancy/awareness claims; benign engineered proxies; no unaware-operator condition in MVP
Next step	Architect review → SPLD carve-out decision → Tier 0 Python stack → review-group card pass → gated run

No AI collaborator should treat any section of this document as finalized, canonical, or ready for downstream development until the Architect applies ONESTAMP. Author-input items and review-group locks from v0.3 remain hard blocks that no model can resolve unilaterally.

Selected sources informing the research account (not an exhaustive bibliography): Nyquist–Shannon sampling theorem (standard DSP treatments); Purves, Paydarfar & Andrews, "The wagon wheel illusion in movies and reality," PNAS 93(8), 1996; Finlay & Dodwell, Perception & Psychophysics 41, 1987; Martineau, Aguilar & Glass, Phys. Rev. Lett. 103, 2009 (predicting wagon-wheel perception); Eagleman & Sejnowski, "Motion integration and postdiction in visual awareness," Science 287, 2000; Patel, Ögmen, Bedell & Sampath, "Flash-Lag Effect: Differential Latency, Not Postdiction," Science 290, 2000; subsequent generalized/cross-modal flash-lag studies (Scientific Reports, 2019; NCBI PMC6405767); distributed-systems causal-time practice (Lamport timestamps, vector clocks, hybrid logical clocks); NTP/PTP operational practice. Internal baselines: Open Window Decision Record v0.3 (2026-08-14); Open Window Implementation Memorandum v0.3; OAL-CFTRL-NAIL-001 v0.1-PPLX; Presentation-State Integrity Framework Independent Development v1.0; PSIF Copilot Independent Pass v1.0; OAL-TA-001 v0.1-GROK.